Akoizo Original Practice - Form 2

Division C | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) In a typical Boomilever, which member is in tension?

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 15 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. The Loading Block

B. Every member equally

C. The top chord that hooks on the bolt

D. The strut pressing on the wall

Answer: _____

2. (2 points) A strut bowed sideways and then snapped in the middle. What is the best next change?

A. Add a brace at mid-length

B. Use thicker chord wood

C. Add glue to the hook

D. Increase the reach

Answer: _____

3. (2 points) Before sand is added, the bottom of the Loading Block must be:

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. Above the horizontal plane through the bolt centerline

B. Below the bolt plane

C. Touching the wall

D. Exactly 45 cm from the wall

Answer: _____

4. (4 points) Using T = W x L ÷ d, find the chord force for 147 N at L = 0.40 m and d = 0.15 m.

Answer: _____
_____
_____

5. (4 points) Why might a team choose the base option even though the bonus is large?

Answer: _____
_____
_____

6. (4 points) Why should a team build a practice wall matching the official dimensions?

Answer: _____
_____
_____

7. (4 points) Explain why a box or spaced-pair strut beats a solid square of the same mass.

Answer: _____
_____
_____

8. (4 points) Why does a flatter strut need to push harder?

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Answer: _____
_____
_____

9. (4 points) How can slow-motion video help failure analysis?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

Answer: _____
_____
_____

10. (4 points) Why must the hook fit within the bolt's 3.0 cm clearance?

Answer: _____
_____
_____

11. (2 points) Define "Reach" in the context of Boomilever.

Answer: _____
_____
_____

12. (2 points) Define "Tier 2" in the context of Boomilever.

Answer: _____
_____
_____

13. (2 points) What most likely caused the difference?

Supplied synthetic observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

A. Different wood in Run 2

B. The bonus line

C. Rushed loading and bucket swing in Run 1

Answer: _____

14. (2 points) Define “Density” in the context of Boomilever.

Answer: _____
_____
_____

15. (2 points) Define “Compression member” in the context of Boomilever.

Answer: _____
_____
_____

16. (2 points) Define “Controlled variable” in the context of Boomilever.

Answer: _____
_____
_____

17. (2 points) Define “Estimated Load Supported” in the context of Boomilever.

Answer: _____
_____
_____

18. (2 points) Define “End grain” in the context of Boomilever.

Answer: _____
_____
_____